

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Final

M.C.A. (Monsoon Semester-2022)

Course Name: Computer Architecture

Maximum Marks: 60

Course Code: CS 405

Total Time: 3 hours

- Q1.** a) What is the difference between X86 and ARM type of Computer Architecture?
b) What are the major factors (performance parameters) that can help in improving the CPU performance in a Computer System?
c) Three enhancements with following speedups are proposed for new architecture:
Speedup₁=25; Speedup₂=15; Speedup₃=10; only one enhancement is useable at time
(i) If enhancement 1 and 2 are each usable for 21% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 15?
(ii) Assume the enhancement can be used 20%, 35% and 18% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?

(3+3+4)

Q2. a) A non pipeline processor has a clock frequency of 300 M Hz and an average CPI of 4. A better processor Y is designed with a 5 stage instruction pipeline. However to latch delay and clock slew the clock rate of Y is 250 M Hz. If a program consisting of 5000 instructions are executed by both the processors, what will be the speed up of processor Y with respect to processor X.

- b) For the following reservation table
(i) What are the forbidden latencies
(ii) Show the state transition diagram
(iii) List all the simple and greedy cycles
(iv) Determine the optimum MAL
(v) Determine the pipeline throughput with $\Delta = 10\text{ns}$

	1	2	3	4	5	6
S1	X		X		X	
S2		X		X		X
S3			X		X	

	1	2	3	4	5
S1	Y			Y	Y
S2		Y	Y	Y	
S3	Y	Y		Y	

(3+7)

Q3. a) What is the difference between 1 cell SRAM and DRAM. Which type of memory is used at various levels of memory Hierarchy Design and why?

b) Consider the pipeline stage with

	Load	Store	R type	Branch	Jump
Instruction Mix	20%	15%	40%	15%	10%

Assuming memory access is 200 ps, ALU operation is 50 ps, register file read/write is 20 ps, half of the load instruction is immediately followed by an instruction that uses the results of branch delay on mis-prediction on one clock cycle and $\frac{1}{4}$ of the branches are mis-predicted. Jumps pays one full clock cycle of delay. Calculate CPI, cycle tie and target address in ID stage to fetch the next instruction so that average time in pipe is 2 clock cycle. (For pipelining average load =1.5 clock cycles; store =1 load cycle; R-type =1clock cycle; jumps =2 clock cycle; average branch =1.5 clock cycles)

(4 +6)

Q4. What are the different ways to improve cache performance.

(b) Consider a CPU with an average CPI as 2.2. Assume the instruction mix is ALU-50%, Load-15%, Branch-35%. Assume the cache miss rate of 2.1% and miss penalty of 50 cycles ($=t_{MM}$). Calculate the effective CPI for a unified L1 cache, using write back and write allocate, with the cache block being dirty is 12%.

(3 +7)

Q5. a) What is Flynn classification for Computer Systems?

b) Discuss vector processing. Consider the multiplication of two 30 X 30 matrices using vector processor. How many product terms are there in the inner product and how many inner products must be evaluated?. How many multiply add operations are needed to calculate product matrix.

c) Suppose start up time for vector multiply operation is 15 clock cycles. After start up, initiation rate is 1.5 per clock cycle. What will be the number of clock cycles required per result for a 64 element vector?.

(3 +5+2)

Q6. a) Explain various types of hazards in the pipelining technique?. What is stalling?.

b) A weather forecasting computation requires 300 billion floating point operations. The problem is processed in a Super computer that can perform 100 megaflops. How long will it take to perform these calculations?

c) Explain arithmetic pipelining.

(4 +3+3)